

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Digital Signal Processing	Course Code:	EE3031
	Program:	BS Electrical Engineering	Semester:	Spring 2025
	Assignment Submission Date:	17-Feb 2025	Total Marks:	32
	Section:	BEE-6A	Weight:	3
	Exam Type:	Assignment- 1	Page(s):	2
			CLO #	2

Student Name: _____ **Roll No.** _____ **Section:** _____

Instructions: 1. Submission instructions depending on whether it is a hardcopy/softcopy submission
2.

Question No. 1 (CLO No. 2)

Marks: 14

- a) Apply transformation on the following system $y(n)$ to determine whether it is time invariant or time variant
- $$y(n) = T[x(n)] = x(n^3)$$

To clarify the result in part (a) assume that the signal $[x(n)]$:

$$x(n) = \begin{cases} 2, & 0 \leq n \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

is applied into the system.

- Sketch the signal $x(n)$.
- Determine and sketch the signal $y(n) = T[x(n)]$.
- Sketch the signal $y'_2(n) = y(n + 3)$.
- Determine and sketch the signal $x_2(n) = x(n + 3)$.
- Determine and sketch the signal $y_2(n) = T[x_2(n)]$.
- Compare the signals $y_2(n)$ and $y(n + 3)$. What is your conclusion.

Question No. 2 (CLO No. 2)

Marks: 10

A discrete-time system can be

- Static or dynamic
- Linear or nonlinear
- Time invariant or time varying
- Causal or noncausal
- Stable or unstable

Apply the following transformations with respect to the above properties.

- $y(n) = \cos[x(n)]$
- $y(n) = \sum_{k=-\infty}^{n+1} x(k)$
- $y(n) = x(n) \cos(\omega n)$
- $y(n) = -x(n + 2)$
- $y(n) = |x(n)|$
- $y(n) = \text{sign}[x(n)]$

Perform and plot the convolutions $x(n) * h(n)$ and $h(n) * x(n)$ for the pairs of signals shown in Figure 1.

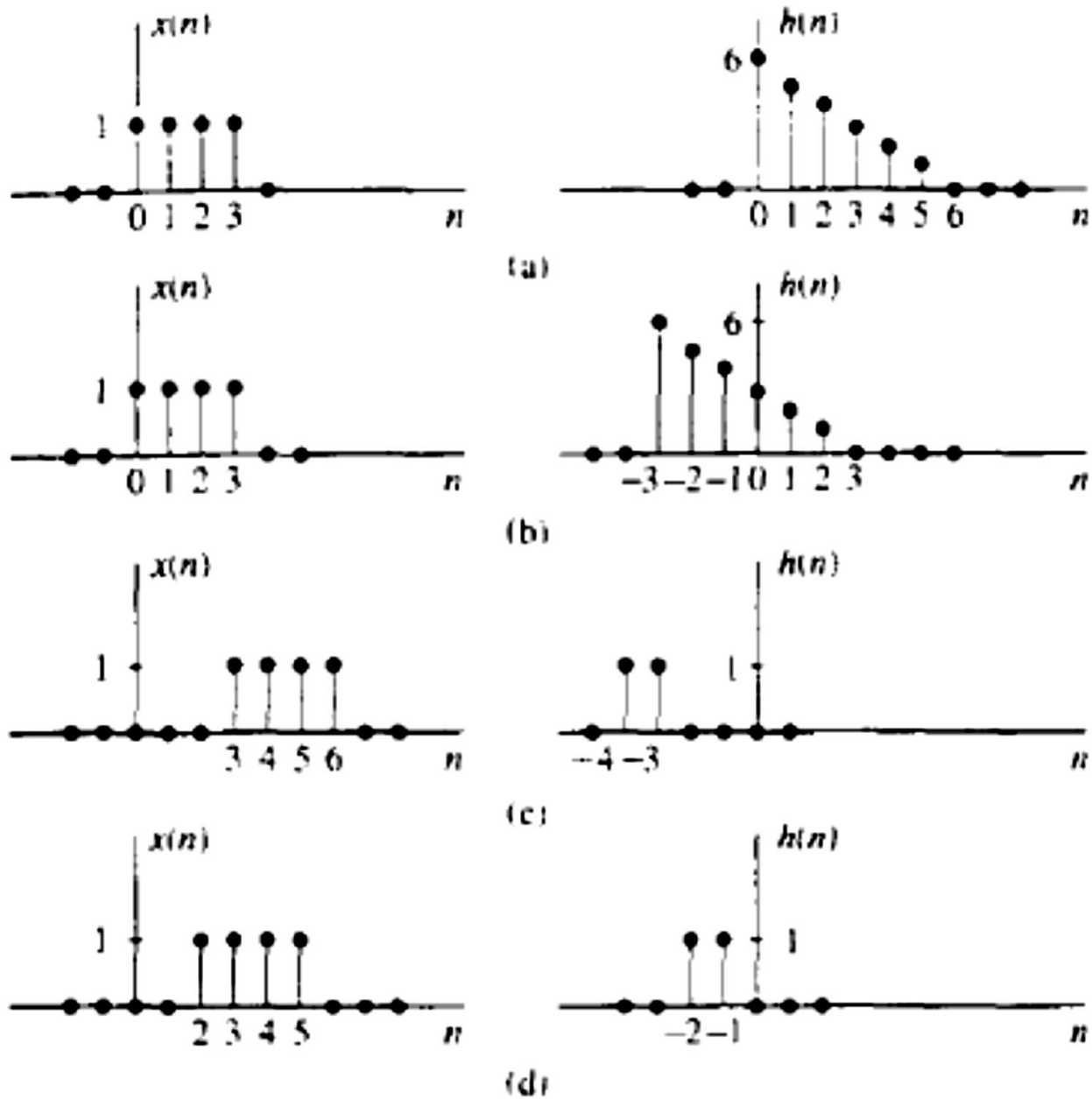


Figure 1